

Proposed Action Plan for Rejuvenation of River Kabini



Karnataka State Pollution Control Board

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INDEX

Sl. No.	Topic	Page No.
1	Introduction to Kabini River	3
2	Sources of Pollution - Municipal Sewage generation and Treatment	4 - 6
3	Characteristics of River water quality	6-7
4	Action taken by the Board	7
5	Action to be taken for Rejuvenation of River Water Quality	7-8
6	Cost component involved in the Restoration of Polluted stretch	8-9
7	Status of Environmental Flow (E-Flow)	9
8	Short Term and Long Term Action and the Identified Authorities for initiating actions and the time limits for ensuring compliance	9-14

Proposed action plan for Rejuvenation of River Kabini

07. State : **Karnataka**
River Name: **Kabini**
River Stretch: **Nanjangud to Hejjige**
Priority: **IV (BOD 6-10 mg/L)**
BOD Range: **3.6-6.5 mg/L**

1.River Kabini:

The Kabani also called Kabini or Kapila, is one of the major tributaries of the river Cauvery in southern India. It originates in the Wayanad District of Kerala state by the confluence of the Panamaram River and the Mananthavady River. Kabini river takes birth at Kerala State. Kabini River joins Cauvery at T.Narasipura, Mysuru District. The total stretch of the river Kabini extends to about 105 Kms. CMC Nanjangudu is the only local body located on the bank of the river Kabini. **The total polluted stretch of the river is about 9Kms i.e., from Nanjangud water supply intake point to Hejjige Village.**



Figure 1.The Kabini River Stretch

2. Major Polluting sources :

2.1 Discharge of sewage from Local Bodies from Nanjangud:

The Population of CMC Nanjangud town as per 2011 census is 50,564. The water supply is around 10.9 MLD. The generation of sewage is estimated as 8.72 MLD. **About 40% of the town is only provided with UGD facility.** The sewage and sullage from the area which is not provided with UGD and other missing links are discharged in to the river Kabini. .

2.2 Dumping of Municipal solid wastes on the banks of river from Nanjangud:

The quantity of Municipal Solid waste generated per day is about 20 Tons / day. MSW site has been identified and necessary infrastructures facilities to process the wastes are provided. At present they are not segregating the wastes and the wastes is being dumped in MSW site. Apart from dumping the solid wastes in the MSW site, some quantity is also dumped on banks of the river.

2.3 Vehicle Washings by the publics at different places.

2.4 Non point sources from Agricultural fields / Farms etc.

2.5 Status of Industrial effluent and treatment facilities:

More number of Industries are highly concentrated in KIADB Industrial AREA and Thandya/ Adakanahalli industrial area of Nanjangudu Taluk. Namely Sugar Industry, Distilleries, Steel, Paper, Pharmaceutical, food processing, textile, etc & the list of industries is given below in the table. All the industries in the jurisdiction of Nanjangudu taluk have provided effluent treatment plant/ sewage treatment plants for the treatment of trade/sewage effluent generated from the industrial activity/ domestic activity & the treated water is completely utilized into on land for gardening within the Industry premises. Hence No industries in the Nanjangudu taluk are discharging the effluents generated from the domestic activity/industrial activity into the river

ACTION PLAN FOR REJUVENATION OF RIVER KABINI

5

The category wise number of industrial units & treatment units provided for the units are given in the below table.

Category	Large	Medium	Small	Total	Details of Major Industries	Mode of treatment	Status
Red	19	3	8	29	Beer Manufacturere – 1No,Organic chemicals- 2Nos,Pharmaceuticals - 2Nos, PCB – 1Nos, Pulp& Paper – 4Nos, Sugar & Distileries- 1No,Sythetic fibre- 1No, Textiles-3Nos., other miscellaneous- 14Nos	All the industries in the jurisdiction of Nanjangudu taluk have provided effluent treatment plant/ sewage treatment plants for the treatment of trade/sewage effluent generated from the industrial activity/ domestic activity & the treated water is completely utilized into on land for gardening within the Industry premises	No industries in the Nanjangudu taluk are discharging the effluents generated from the domestic activity/industrial activity into the river .
Orange	7	0	10	17	Potable alchohal-1, Instant coffee – 1No, Steel- 1No, Food processing – 1, Solvent Extraction -1, Tobacco Processing- 1, other miscellaneous- 10Nos		
Green	5	1	78	84	Granite cutting- 35, Pulverizer-3, Rice mill- 9, Stitching units- 9, other miscellaneous- 28Nos		
White	1	1	8	10	Miscellaneous- 10Nos		

2.6 Drainout-falling in River Kabini:

There are no drains discharging into River in the surroundings of the Nanjangudu. However Sewage treatment facility is not provided to treat sewage effluent. The sewage and sullage from the area which is not provided with UGD and other missing links are discharged in to the river Kabini.

2.7. Municipal Sewage generation and Treatment:

The Town wise sewage generation and treatment capacity developed so far is provided in **Table-1**

Table-1: Status of Domestic Pollution in River - Kabini

Sl No.	Name of the local body	Type	Total Sewage generation in MLD	Total Capacity of Sewage treatment in MLD	Status of STP
1	Nanjungud	CMC	8.72	7.0	Operational at present

3. Characteristics of River water quality:

The monitoring results of **Kabini** River At Bathing Ghat, Nanjungud & Water supply intake point to T.Narsipura for the year 2017 & 2018 are shown in **Table-2** River water quality conforms to Class D- means that the water is fit for propagation of wildlife, fisheries at (Bathing Ghat) for the year 2017 & for the year 2018 the same station conforms to class E-means that the water is fit for irrigation, industrial cooling and controlled waste disposal .For the years 2017 & 2018 river water quality of Water supply intake point to T.Narsipura conforms to Class C-means that the water is fit for drinking purpose after conventional treatment followed by disinfection.

3.1 Status of Water Quality

The details of parameter and specific concentration are provided in **Table-2**

Table-2 : Status of Water Quality of River - Kabini

Year	Locations	DO (mg/L)		BOD(mg/L)		Fecal Coliform (MPN/100ml)		Total Coliform (MPN/100ml)		Class
		Min	Max	Min	Max	Min	Max	Min	Max	
2017	At Bathing Ghat,Nanjangud	4.1	6.1	2.4	6.5	320	790	1700	2400	D
	Water supply intake point to T.Narsipura	4.8	7.1	1.2	3.0	210	460	1100	2100	C
2018	At Bathing Ghat,Nanjangud	3.0	7.0	1.06	4.1	320	840	1500	2200	E
	Water supply intake point to T.Narsipura	6.8	7.0	1.04	2.0	140	330	1100	1300	C

The results indicate that the water is polluted due to sewage from Nanjangud Town.

4. Action taken by the KSPCB:

1. New water quality monitoring stations in the stretch based on the criteria like upstream / down stream of confluence point of tributaries, sewage, agriculture run off etc have been established.
2. CMC , Nanjangud has been informed to provide UGD and STP to the whole municipal area and to treat to the standards prescribed by the Board and to make use of the treated sewage for irrigation / secondary purposes. In this regard, they have been advised to submit action plan for the treatment of sewage and utilization of which includes solid waste management.

5. Actions plan to be taken for Rejuvenation of River water quality:

- The CMC Nanjangud shall provide 100% UGD system to the entire town.
- The CMC Nanjangud shall provide terminal sewage treatment plant to treat the sewage to the standards stipulated by the CPCB.

- The CMC Nanjangudu shall stop discharge of un-treated / treated sewage directly in to the river. Treated sewage shall be utilized for irrigation purpose. Un-treated sewage shall be used for farming till full pledged STP is established.
- Municipal solid wastes including poultry and slaughter house wastes shall be disposed only in MSW sites as per MSW guidelines and shall not be disposed on river banks.
- Vehicle washings and other activities shall be regulated.
- There shall not be discharge of sewage or sullage in to storm water drains. If any discharges are there same may be regulated by taking measures not to discharge into river.
- River cleaning and rejuvenation activities like removal of algae, sediments, weeds etc., shall be undertaken.
- Caution display boards at strategic points shall be displayed.
- Creation of buffer zones on either side of the river bank and developing green belt in that area. No developmental activities shall be taken in this area.
- Creation of buffer zones on either side of the river bank and developing green belt in that area.
- Construction of bathing ghat at Nanjangud by temple authority.
- Banning of dumping of clothes and waste into the river by devotees by temple authority.

6.0. Cost component involved in the Restoration of Polluted stretch

Cost component shall be an integral part of Detailed Project Report (DPR). Most of the cities and towns are deficient in treatment of its total sewage generated. In order to cater each identified town on the bank of polluted river and gaps observed between total sewage generated and treatment capacity needs to be considered for planning.

Cost component shall invariably depend towards construction, operation and maintenance of sewage treatment plant. On an average Rupees 2.5 Crore has been estimated as Capital Cost per MLD (for primary, secondary and Tertiary treatment) excluding Operation and maintenance cost for all the available conventional and recent technologies. In some cities and towns developed capacity of STP is fully or partially underutilized due to inadequate sewerage network and other implementation issues.

Total estimated cost of **Rs. 80 Crores** should be made budgetary provision for Capital cost including O&M for existing STP / proposed new STP's in the identified city along the Kabini River.

Table -3: Cost Component involved in the Rejuvenation of Polluted Stretch of Bhadra

Sl. No.	Activity	Cost in Crores
		Nanjungud
1	Operation & maintenance (O&M) cost for existing STP per annum (7 MLD) including UGD facility for 60% remaining uncovered areas & to provide HSCs to entire city.	1.18 Crores /annum 78.82 Crores (UGD-60%)
	Total Rupees	80 Crores

7. Status of Environmental Flow (E-Flow) :

The details of Flow (discharge) is provided in **Table-4**

Table-4 : Status of E-Flow of River - Kabini

Year	Hydrological Observation Site	Flow (m ³ /s)	
		Min	Max
2015	T.Narsipura	7.688	388.428
2016		8.928	37.684

8. Action Plan- Short Term and Long Term Action and the Identified Authorities for initiating actions and the time limits for ensuring compliance.

Short term and long term action plans and the implementing agencies responsible for execution of the action plans and the time limits are given in table as below :-

ACTION PLAN FOR REJUVENATION OF RIVER KABINI

10

Sl. No	Action plan for rejuvenation of river Kabini	Organisation/ Agency Responsible for Execution of the Action plan	Time Target
I.	Industrial Pollution Control		
	(a) Compliance of industries located in catchment area with respect to effluent discharge standards and its disposal as per consent conditions	KSPCB	Complied
	(b) Inventorisation of the industries in the catchment area of River KABINI covering assessment on aspects relating to Status of Consents under Water & Air Acts and Authorisation, Effluent Generation, ETP capacities and final mode of effluent discharges	KSPCB	Complied
	(c) Actions against the Identified industries in operation without Consents under Water & Air Acts/Authorisation under the H&OW (M & TM) Rules, 2016 as amended	KSPCB	Complied
	(d) Action against the industries not installed ETPs or ETPs exist but not operating or ETP outlet or treated effluent is not complying to the effluent discharge standards or norms	KSPCB	Complied
	(e) Action against the red category industries for installation of OCEMS and not transferring data to CPCB and KSPCB	KSPCB	Complied

ACTION PLAN FOR REJUVENATION OF RIVER KABINI

11

	(f) Small scale/tiny and service providing units located in urban or semi-urban limits like Dairies, Auto Service Stations to have minimum provision of O & G traps	Local bodies(Nanjungud)/DMA	Within three months
	(g)Prohibition of Burning of any kind of waste including agro-residues	State Govt./District Administration and Local bodies(Nanjungud) and agriculture dept.	Within three months
	(h) Directions to all the Industries which are observed to be not in operation or closed or temporarily closed to remain close till further orders fromCPCB.	KSPCB	Within three months
	(i) Estimation of industrial effluent generation and the existing CETP capacity and to arrive gap between the industrial effluentgeneration and the existing treatment capacity	KSPCB	Not Applicable
	(j) Channelization of industrial effluents to CETPs for ensuring treatment to comply with the discharge standards.	KSPCB and District /Local Administration	Not Applicable
	(k) Identification of suitable site within industrial estates, Execution and Commissioning of Adequate Capacity CETPs.	State Government , District/Local Administration /KIADB	Not Applicable
II.	Sewage Treatment and Disposal Plan		

ACTION PLAN FOR REJUVENATION OF RIVER KABINI

12

	(a) District-wise estimation of total sewage generation, existing treatment capacities, quantum of disposal of sewage presently through drains and the gaps in sewage treatment capacity.	State Government, KUWS & DB, District Administration/ Local bodies(Nanjangud)	Within six months
	(b) To undertake measurement of flow of all the drains presently contributing pollution load in river KABINI and to formulate detailed project report (DPR) for each drain and corresponding town and submission of DPR.	State Government, KUWS& DB, District Administration & Local bodies(Nanjangud)	Within six months
	(c) Proper design, execution of STPs with full utilisation capacity	State Government, KUWS& DB, District/Local Administration	Within 24 months
	(d) Channelization including diversion of sewage generated from household/town ships/villages to sewer lines/interception of all the drains presently carrying sewage and for ensuring proper treatment through the upcoming STPs	State Government, KUWS & DB, District/Local Administration	Within 24 months
	(e) Ensuring dairy/automobile service stations and Hotels / Restaurants particularly located on road-side should have a treatment system and levy of fine in case found violations	Local authorities/DMA	Within three months
III	Ground water quality		
	(a) Sealing of contaminated hand pumps and found to be unfit for drinking purpose by the public	State Government, Karnataka rural drinking water and Sanitation Department and Local authorities	Contaminated ground water is not noticed

ACTION PLAN FOR REJUVENATION OF RIVER KABINI

13

	(b) Supply of potable water to the affected communities in the identified critical blocks	State Government, Karnataka rural drinking water and Sanitation Department and Local authorities	Not Applicable
	(c) Carrying assessment of ground water survey for quality and to identify over exploited and critical blocks in the district (Mysuru.).	Karnataka Ground Water Authority	Within Six months
	(d) To conduct periodic surprise inspection of the industry to rule out any forceful injection of industrial effluents into ground water resources	KSPCB/KGWA	Complied
	(e) All the industry should be directed to obtain NOC from the CGWB and action against the Units in Operation without obtaining of NOC from CGWA	KSPCB,CGWB/CGWAand KarnatakaGround Water Department	Within six months (The proposed new industries will be directed to obtain NOC from CGWA)
	(f) To ensure rain water harvesting by the industrial, commercial and other institutions and groundwater recharging with only clean water be encouraged by CGWB/CGWA	CGWA/ Karnataka Ground Water Department	Within six months

IV	Flood Plain Zone (FPZ)		
	(a)Plantation in Flood Plain Zone (FPZ)	Karnataka State Forest Department	Within Six months
	(b)Checking encroachments in the FPZ of river Kabini	District/Local administration	Within Six months
	(c)Prohibition of disposal of municipal plastic and bio- medical waste particularly in drains	Local administration	Within Six months

ACTION PLAN FOR REJUVENATION OF RIVER KABINI

14

	(d)Notification of Flood Plain Zone FPZ	State Government	Within six Months
V	Environmental Flow (E-Flow) and Irrigation Practices		
	(a)Measurement of flow in the river and records maintained	Central water Commission/Karnataka Irrigation Department/Water Resources Department	Regularly (Daily/monthly)
	(b) To conserve water and good irrigation practices to be adopted by the farmers by organising mass awareness programmes and through media in vernacular language	Karnataka water resources department/Karnataka .Irrigation Department.	Once in six months
